Preckon

An AI-native construction operating system where every number can name the document, the drawing layer, or the person it came from.

The most consequential number your business produces is the least governed one.

A tender price commits the company for years. It is produced in **two to six weeks**, by one senior estimator, in spreadsheets, from marked-up PDFs — and when it is questioned in a post-tender meeting, the answer comes from memory.

CONCENTRATION

The method lives in one person's head. It leaves when they do.

OPACITY

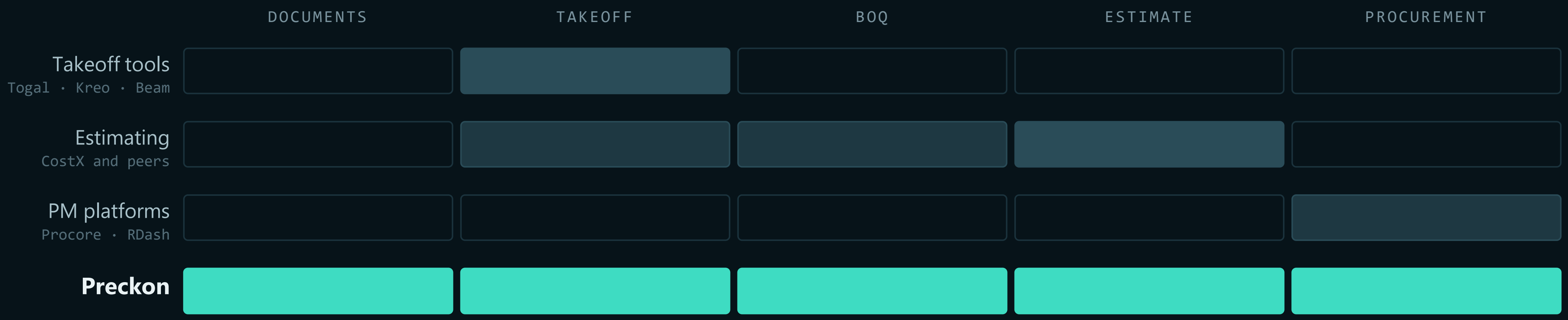
No trail from a number back to the drawing it was measured off.

LATENCY

The cycle is longer than the bid window, so scope gets thinner.

Every other business-critical number you hold — payroll, ledger, inventory — sits in a system with an audit trail. The estimate does not.

The preconstruction chain has five stages. Almost every tool owns one.



> *Everyone sells a stage. The cost, the delay and the lost provenance are all in the joins between them.*

SOLID = OWNS THE STAGE · MID-TONE = DOES IT, BUT MANUALLY · OUTLINE = DOESN'T REACH IT

Not a takeoff tool. Not an estimating product. Not a PM clone.

Preckon is a construction operating system: one continuous chain from raw tender document to issued procurement package, with a governed AI fabric sitting between every module and every model.

01 **Connect the lifecycle.** Project information stops being trapped in drawings, spreadsheets, email and departmental tools.

02 **Turn project data into intelligence.** Deterministic construction engines first, structured project memory second, governed AI third — in that order, always.

03 **Adapt to scale.** A complete native platform for mid-market, and an integration-and-intelligence layer for enterprises that cannot replace ERP, P6 and Autodesk on your timetable.

TENDER	DRAW	DOC	QUANT	COST	SCHEDULE	PROCURE	NARRATIVE
Reads the package. Extracts requirements, structures the bid.	Interprets PDF and CAD. Recognises elements, measures them.	Control, revisions, transmittals. Ties clauses to what they govern.	Takeoff and the bill, measured from the above.	Rates, assemblies, the priced estimate.	Deterministic CPM, calendars, baselines that refuse silent moves.	Packages grouped from the bill, ready to issue.	The technical submission, from the same records.

Each is not a screen — it is an intelligence layer over the same project record. Nothing is retyped between them, because nothing is ever *handed over*; the next stage reads the confirmed output of the last.

All eight are seeded and running in the deployed workspace today. `Preckon Core` underneath carries identity, tenancy, workflow, entitlements, audit and the AI gateway — modules are forbidden from re-implementing any of it.

Every number on screen can tell you where it came from.

Which page of which document. Which layer of which drawing. Or which person typed it, and when they signed for it.

> *That is the product. The speed is a consequence of it, not the point of it.*

The commercial model is data. The product plane never learns your contract.

HOST PLANE — CONTROL

Where the product is defined

Feature → Edition (with limits) → Pricing → Subscription → **resolved entitlements**. Tenant provisioning, billing, platform operations.

`32 tables · append-only hash-chained audit · operated by TechSME`

TENANT PLANE — WORK

Where the project lives

Projects, documents, drawings, artifacts, review queues, the audit chain. Consumes exactly **one** thing from the host: resolved entitlements.

`84 tables · 90 API routes · deployed`

Because the split is that clean, a tenant can be given its own deployment target, its own region, or its own brand without touching product code — and pricing can change without a release.

The core knows nothing about construction.

It knows one interface: `DomainPack`. A vertical is data — one file satisfying the contract, plus one line in the registry.

A pack declares its modules, artifact types and JSON schemas, agents with typed I/O, workflows as DAGs, personas, a lifecycle state machine, a role template, permissions and standard rules.

CONSTRUCTION PACK — SHIPPED

8 modules · 18 artifact types · 20 agents · 14 workflows · 34 deterministic engines

THE PROOF IT IS REAL

A second pack — **underwriting** — runs on the same unmodified kernel. Construction is not hard-coded; it is loaded.

THE ABI — FOUR SYSCALLS, NOTHING ELSE

`emitArtifact` · `readArtifacts`

`enqueueJob / onJobResult` · `requestReview`

Every output is a typed record that carries its own parentage.

WHAT PRODUCED IT

The agent, the model alias, the prompt version, the run.

WHAT IT DERIVED FROM

The upstream artifacts, the document page, the drawing region.

WHO CONFIRMED IT

The named person, the timestamp, the entry on the audit chain.

AND THE PART NOBODY ELSE DOES

Correct a confirmed record and the system keeps your version, records the disagreement against your name, and **marks every downstream record stale** — then tells you precisely what needs re-running. Not the correction. The consequences of the correction.

AI does not own authoritative business state.

Architecture principle P-02, locked as `ADR-003`. Not a setting. Not a policy document. A structural property of the ABI.

AI PRODUCES

Proposals, extracted facts, classifications, draft artifacts, explanations — each with confidence and provenance attached.

AI CANNOT COMMIT

A BOQ change. A schedule baseline. A contract value. A drawing revision. An approval. Any authoritative transaction.

Confirming is a **permission**, held by a person, checked at a choke point. Every confirmation lands on the tamper-evident audit chain against a name and a time.

No module ever calls a model.

1	**Deterministic software**	DEFAULT
2	**Exact and semantic cache**	BEFORE INFERENCE
3	**Structured retrieval**	BEFORE LONG CONTEXT
4	**Small private model**	CLASSIFICATION
5	**Private reasoning model**	EXTRACTION
6	**Frontier model**	ESCALATION ONLY

MODEL INDEPENDENCE

Code references aliases — `preckon-reasoning` — never a vendor or a model name. Swapping providers is a registry change.

POLICY BEATS CAPABILITY

A restricted document does not leave its approved boundary because an external model happens to be better at the task.

EVERY CALL HAS ECONOMICS

Attributed per attempt to tenant, project, user, module, task, route, tokens and cost. AI spend is a product KPI, not an invoice surprise.

AUDIT

Hash-chained

Append-only, tamper-evident, and verifiable from the interface — one button, in front of the customer.

ISOLATION

One choke point

Every tenant-scoped query passes a single enforced path. A cross-tenant read returning zero rows is an assertion in the test suite.

PROVENANCE

On every derived value

Source, version, rule and model recorded at the moment of derivation, not reconstructed later.

DURABILITY

State in the database

Jobs, gates and re-plans are durable rows, not in-memory queues. A restart resumes; it does not lose the run.

LIVE WORKSPACE

Five minutes on the running system.

app.preckon.com — a real tenant, real documents, nothing pre-rendered

BEAT 01 · 60S

Propose

Open a live bid. Run a stage. Agents read the tender and the drawings and produce proposals — nothing is committed.

BEAT 02 · 60S

Dispose

The review queue. Confirm one, reject one. Watch the bid pursuit advance because a human decided, not because a card was dragged.

BEAT 03 · 120S

Correct

They pick a BOQ line. Trace it to the drawing layer it was measured from. Then disagree with it — and watch the downstream go stale.

BEAT 04 · 60S

Verify

Trace view, then Verify audit chain — live, in front of them. Then switch the workspace to Arabic and leave it for a beat.

132 features. 97 done, 15 partial, 20 missing.

Measured against the blueprints *as written* — 13,074 lines of specification across ten documents. Not against a roadmap we could move.

HOW IT IS COUNTED

Done means built, tested and deployed. Partial counts as a half. Missing counts as zero. Every row was verified against the codebase and the deployed production database.

84	database tables, deployed
90	API routes
982	tests passing, 46 files
34	deterministic engines
24	schema migrations
20	AI agents, typed I/O
18	artifact types
14	workflows
12	architecture decision records
31	live sheets in production

These were queried from the repository and the running production database on the morning of this briefing. Any of them can be re-run in front of you.

ENTERPRISE REQUIREMENT	STATE	NOTE
SSO — SAML / OIDC	NOT BUILT	Deal blocker. Named in the platform spec.
MFA	NOT BUILT	Deal blocker.
Encryption at rest + KMS	NOT BUILT	Specified, not yet implemented.
CI running the suite	0 RUNS	Six workflow files exist; none has executed.
Staging environment	NONE	Changes are validated in production.
Backup with tested restore	NONE	An untested restore is not a backup.
Retrieval / RAG	PARTIAL	Wired for DocLogix; no embeddings generated yet.
AI policy enforcement	REPORT-ONLY	Wired and recording; enforcement behind a flag.

Sixteen open P0s. These are the eight you would have found.

You will find every one of these in a diligence pass. We would rather you heard them from us, in the room, than from a consultant in week six.

The trust foundations underneath — audit chain, tenant isolation, provenance, durable jobs — are built and tested. What is missing is the enterprise *operations* layer around them. That is a known, bounded, unglamorous list.

SHAPE 01

Preckon SaaS

Multi-tenant, managed. Fastest to value, standard commercial terms.

SHAPE 02

Dedicated cloud

Your region, your isolation boundary, our operations.

SHAPE 03

Private AI

Inference inside your boundary. No document reaches an external provider.

SHAPE 04

Sovereign AI

Full on-premise, including the model serving layer, for regulated and state work.

There is no enterprise fork. The same domain model, identity model, API contracts, workflow concepts, audit model and AI governance operate across all four. The deployment shape is configuration and hosting — never a second codebase to maintain, and never a second product to certify.

Commercially the same fabric is presented as three capability tiers — `Edge`, `Construction AI`, `Frontier` — which are packaging over one architecture, not three of them.

PHASE A · NOW

Coexist

Import and export against the Autodesk ecosystem, IFC, DWG and DXF, PDF, Excel, P6 and your ERP. Preckon adds a layer; nothing is switched off.

PHASE B

Reduce dependency

Takeoff, BOQ, tender workflow, document control and commercial workflows move natively, one at a time, on evidence.

PHASE C

Native

Preckon becomes the system of work from design through handover — while open exchange with consultants and owners is retained permanently.

> *We are not asking anyone to migrate. Phase A has to earn Phase B.*

Three things, six weeks, one live tender.

01 **One real bid, end to end.** One project, one estimating team, six weeks — measured against a bid you have already priced, so the comparison is yours and not ours.

02 **A golden set.** One design set paired with its final priced BOQ, which we hold as a regression fixture. Accuracy stops being a claim the moment it has a fixture behind it.

03 **A named security counterpart.** Someone on your side who closes that P0 list with us inside the same six weeks — SSO, MFA, CI, staging, backups.

WHAT WE ARE NOT ASKING FOR

A migration. A system to be switched off. A commitment beyond the pilot. Or a decision today.

Twenty minutes, from an empty project to a priced bill, a programme, a technical submission and a register of what is still outstanding.

But the number of minutes was never the argument.

> *Every number you saw today can name where it came from. A page of a tender, a layer of a drawing, or a person who typed it and signed for it.*

That is the whole product. The rest is speed.